

COD Lab: Aug-Dec-2025

UE23EC352A

Lab 2 Expt.2

Addition :

2. ALP to add N , a. 32; b. 16 and c. 8 bit numbers using RV32 ; N=4

Assignment -3

LAB2

d) ALP to implement the expression: $X = (Y + M) - (L - D) + (Z + C) - D$

C,D,M,Y,L,Z stored in memory locations and result to be stored in consecutive memory location-X.

Ex-a: Assembler Directives and register transfer instructions:

.text

li x1,25

li x2,100

add x3,x1,x2

mul x4, x1,x2

li x5,3

add x6,x4,x2

blt x1,x2 next

sw x3,0(x2)

next: j next

Ex-b: Array addition

.data

array: .byte 20,7,10,5,35,40

.text

la x1,array

lb x2,0(x1)

lb x3,1(x1)

add x5,x2,x3

lb x4,2(x1)

lb x5,3(x1)

sub x9,x4,x5

sub x8,x8,x9

lb x6,4(x1)

lb x7,5(x1)

add x10,x6,x7

add x8,x8,x10

sub x11,x8,x5

sb x11,7(x1)

exit:j exit

Expt 2: Write an ALP to add 'N'

a) 32 bit numbers

b) 16 bit numbers

c) 8 bit numbers

(a)

```
.data
num: .word 0xabcdeeff,0x11223344,0xaabbccdd,0x12345678,0x11aa22bb
.text
la x1,num
li x4,5
addi x3,x0,0
rpt: lw x2,0(x1)
add x3,x3,x2

addi x1,x1,4
addi x4,x4,-1
bne x4,x0,rpt
sw x3, 20(x1)
nop
```

Note: We get truncated 32 bit result in above program

Assignment-1: Use 64 bit processor to get upper 32 bit result

(b)

```
.data
num: .word 0x4444abcd,0x1122,0xaabb,0x1234,0xaa22
.text
    la x1,num
    li x4,5
    addi x3,x0,0
rpt: lhu x2,0(x1)
    add x3,x3,x2

    addi x1,x1,4
    addi x4,x4,-1
    bne x4,x0,rpt
    sw x3, 0(x1)
    nop
```

c.

```
.data
num: .word 0xabcd,0x1122,0xaabb,0x1234,0xaa22
.text
    la x1,num
    li x4,5
    addi x3,x0,0
rpt:lbu x2,0(x1)
    add x3,x3,x2

    addi x1,x1,4
    addi x4,x4,-1
    bne x4,x0,rpt
    sw x3, 0(x1)
    nop
```

Assignment-2: Define data as half word in 2b. and data byte in 2c. and modify the programs accordingly.